

Electrolyte Tank

A separate loop feeding the stack’s electrode compartments — never part of the treated-water path

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17.1 Overview and function

The electrolyte tank (tag prefix **TE**) supplies a separate, recirculating loop that feeds the SonixED™ stack's **electrode compartments** (chapter 10) — this is what actually carries current at the electrodes. Unlike the diluate and concentrate tanks above, the electrolyte loop is **not part of the treated-water path**: it never mixes with the feed, diluate or concentrate streams, and is not delivered to the customer. A documented “Electrolyte Preparation” monitoring point (its own conductivity transmitter, distinct from the main process conductivity sensors) confirms the electrolyte solution is prepared and monitored as its own loop, separate from process water quality.



WARNING

Electrode reactions can evolve gases. The system carries a general “Ventilation OFF” alarm (documented warning code W002, appendix A) — while that alarm is a general system warning rather than confirmed as electrolyte-specific, adequate ventilation matters most wherever electrolysis is taking place, i.e. around the stack and electrolyte loop. TODO: HydroVolta engineering should confirm the electrolyte chemistry and any specific gas-hazard/ventilation requirement for this loop — see also chapter 2.

fig-electrolyte-tank.png

Figure 17.1. Suggested figure: photograph of the electrolyte tank and its connections to the stack's electrode compartments, clearly distinguished from the diluate/concentrate piping.

17.2 Key specifications

No finalized standard-product capacity/material specification exists — see chapter 15 for the two real tank orders on file (candidates, role unconfirmed). A documented level-alarm configuration uses narrower thresholds for this tank than for diluate/concentrate — **20 cm (low) / 80 cm (high)** (appendix A) — consistent with this being a smaller support loop rather than a main process buffer, though this has not been confirmed as a general rule.

- Level: TE_Level.
- Electrolyte-preparation conductivity: monitored separately from the main process streams (chapter 13).

TODO: electrolyte solution composition (which chemical, target concentration) and replenishment/dosing method were not found in HydroVolta's documentation reviewed.

17.3 Operation

The electrolyte loop recirculates continuously whenever the stack is powered (chapter 10); level is monitored via `TE_Level` with the narrower alarm band noted above. TODO: no document describes an electrolyte make-up/replenishment procedure, or how often the solution itself is refreshed rather than merely topped up.

17.4 Maintenance

TODO: no maintenance schedule for the electrolyte loop was found. Because this loop directly feeds the electrode compartments, its condition is tied to electrode/membrane health — see the stack maintenance notes in chapter 10 for the (also largely undocumented) membrane/electrode service interval.

17.5 Troubleshooting

HL_Alarm_TE/LL_Alarm_TE cover this tank. An unexpected electrolyte level drop should prompt a check for leaks at the electrode compartment seals (chapter 10) before assuming a sensor fault; verify the sensor itself per section 13.4 either way.